

Inter-Team Coordination Mechanisms in Large-Scale Agile

Helga Nyrud
University of Oslo
Gaustadalléen 23B
0373 Oslo, Norway
helgany@ifi.uio.no

Viktoria Stray
University of Oslo
Gaustadalléen 23B
0373 Oslo, Norway
stray@ifi.uio.no

ABSTRACT

Coordination is an important but challenging success factor in large-scale software development. Large-scale is particularly demanding because it involves several teams, and therefore we chose to study coordination between teams. Our aim of this case study was to investigate inter-team coordination mechanisms. By using an established framework from the field of sociology, we found eleven different coordination mechanisms that we mapped into the five different categories. We interviewed four project members and observed 26 meetings in three different teams as part of a larger digitization project. Our results show that even though there are many impersonal mechanisms in the form of various guidelines and rules, the most important mechanism seem to be ad-hoc conversations between two or more project members. The large-scale project had many regularly scheduled meetings. However, the majority of these meetings tended to be, to a greater extent, about reporting status and not about coordinating work. The scheduled meeting that was the most time-consuming, and also involved little coordination, was the daily stand-up meeting with two teams. Future work should investigate how to increase the value of daily stand-up meetings in large-scale programs, and whether the meetings can be organized in a way that enables more inter-team coordination.

Keywords

large-scale software development, coordination practices, large project organization, daily stand-up meeting, communication, software engineering, multiteam systems

1. INTRODUCTION

An important topic in large-scale agile is inter-team coordination [1][2]. In large-scale settings, achieving effective coordination is difficult because of the complexity different sites and teams and their dependencies introduce, especially when dependencies between activities are tightly coupled [3]

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[4]. Dingsøyr et al. classifies large-scale agile as development projects of two to nine teams where coordination is achieved through a new forum where the teams participate [5].

Agile software development has become the mainstream development methodology [6]. According to a recent survey [7], more than 70% of developers work in agile software projects and almost 40% work in distributed teams. Agile methodology introduced a shift in how teams coordinate their work. For example, most agile teams conduct daily stand-up meetings [8]. Representatives from more than one team often attend daily stand-up meetings to achieve inter-team coordination in large-scale agile and the meetings are often called Scrum-of-Scrums [9].

With large projects, new organizational forms such as "team-of-teams" are being used, referred to as multiteam-systems (MTS). An MTS is defined as a minimum of two teams that work interdependently and are linked by shared super-ordinate goals [10]. Scheerer et al. [11] studied coordination strategies in large-scale agile with an MTS perspective and call for more exploratory research on coordination in large-scale software development.

In this paper, we have conducted a case study that draws on a well-known framework of coordination mechanisms proposed by Van de Ven et al. [12].

We investigate the following research question: *What coordination mechanisms facilitate large-scale agile development?*

2. METHOD

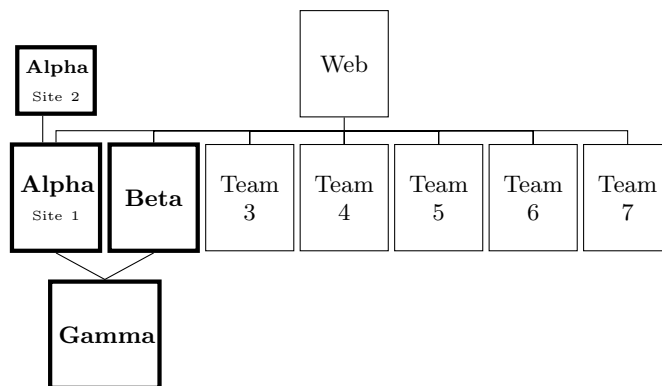


Figure 1: Organizational chart

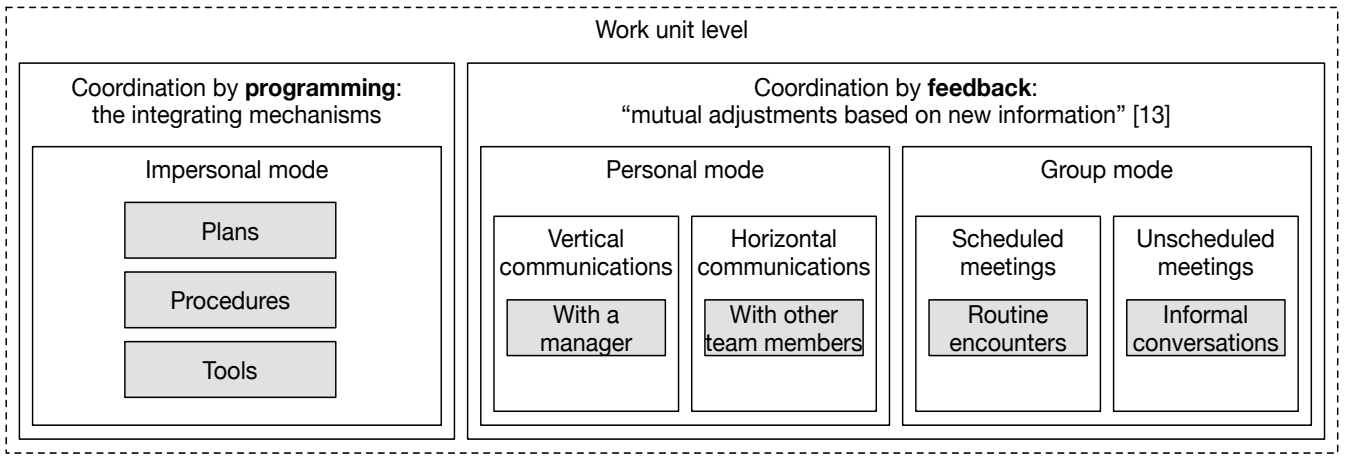


Figure 2: Coordination modes with examples in grey, based on Van de Ven et al. [12]

The study was conducted in a digitization project in a Norwegian pension, bank and insurance company with more than 2,000 employees. The project had eight teams that worked on different solutions, and we studied three of these teams, marked in bold in Figure 1. The three research subjects were two Norwegian teams, Alpha and Beta, and one offshore, outsourced Indian team, Gamma. Additionally, the Alpha team also had an extended team offshore consisting of two developers, named Site 2 in Figure 1.

Team Alpha developed the company’s services for the retail market, while team Beta developed services for the company’s business customers. Both team Alpha and Beta consisted of three developers, one team lead, one tech lead and one test lead. Six of team Gamma’s members were allocated to work on tasks from team Alpha, while three individuals worked on solutions handed to them by team Beta. Gamma was typically handed tasks that required less domain knowledge and understanding of Norwegian law and language. Even though Gamma was an outsourced team with most of their team members offshore, they still had two members on-site, seated close to both Alpha and Beta. All three teams followed agile methodology. The three teams conducted daily stand-up meetings, and we observed 20 of these as shown in Table 1. We also observed six other meetings, such as workshops and team meetings, as well as the project members working in their natural habitat. Additionally, we conducted four interviews, with representatives from team Alpha and Beta. The interviews lasted for approximately one hour and were audio recorded and transcribed. Both notes from observations and the interview transcripts were imported into a tool for qualitative analysis and were coded on the topic of inter-team coordination.

There has been done several studies on coordination mechanisms; hence there are also multiple frameworks on the subject. Thompson [13] proposed the theory that coordination

mechanisms consist of three parts: standardization/rules, plans/schedules, and mutual adjustment. The framework proposed by Van de Ven et al. [12] expanded on the theory by Thompson [13] by adding a fourth part which included the group aspect of coordination, namely the team. This is represented in the rightmost part of Figure 2, and called Group mode. Hence, we chose this framework, because it takes into account the team coordination mechanisms as well.

Furthermore, Mintzberg [14] has a similar theory. His theory states that coordination mechanisms are divided into five: mutual adjustment, direct supervision, and the standardization of work processes, of work outputs, and of work skills. While Mintzberg’s [14] framework is on an organizational level, Van de Ven et al’s [12] framework is on the work unit level of an organization, which makes it more suitable for this study of the inter-team coordination mechanisms in the digitization project.

3. RESULTS

We have mapped coordination mechanisms in the large-scale project according to a framework from classical organisational theory [12], see Table 2. The framework [12] identifies three main coordination categories: impersonal, personal and group mode, depicted in Figure 2. The impersonal mode represents all the forms of coordination by *programming*, meaning the integrating mechanisms such as formalized rules, pre-established plans, standardized information and communication systems [12]. The personal and group mode are both types of coordination by *feedback* and is defined by Thompson [13] as “mutual adjustments based upon new information.” All of the present coordination mechanisms are displayed in Figure 3 according to when they happened during the two-week iteration, in order to present when the different mechanisms took place.

3.1 Programming

The project used five different coordination mechanisms of type programming and impersonal mode, as shown in Table 2.

All three teams followed an agile process. The offshore team Gamma followed the Scrum methodology and conducted two-week iterations (called sprints). The main pur-

Table 1: Data collection

Data	Alpha	Beta	Gamma
Daily Stand-Up Meeting	9	5	6
Other meetings	2	2	2
Interviews	2	2	0

Table 2: Inter-team coordination mechanisms

Coordination Mechanisms	Programming	Feedback			
	Impersonal mode	Personal mode		Group mode	
		Vertical	Horizontal	Scheduled	Unscheduled
Agile process	X				
Jira	X				
Rules for QA	X				
Open work area	X				
Instant messaging	X				
Informal ad hoc conversations			X		X
Stand-Up Meeting				X	
Retrospective				X	
Demo				X	
Sprint planning				X	
Backlog grooming				X	

pose why the off-site team worked with Scrum’s time boxes were to secure predictability as to when their solutions and products were delivered to the Norwegian teams for testing. Thus the use of time boxing enabled coordination because it simplified planning the different coordinating mechanisms. However, the Alpha and Beta teams followed a Scrum methodology where they task boxed their assignments.

The project management tool Jira enabled coordination between different teams. Jira was a crucial part of team Gammas daily stand-up meetings because their meetings were organized so that the on-site Team Lead ran through the tasks that were in the "In Progress" column. Furthermore, they had columns in Jira for Quality Assurance (QA) for the off-site team and for the Norwegian team that owned the technical solution; team Alpha or Beta. Additionally, they had one last column called Acceptance Testing. Moreover, Jira also enabled coordination between the teams by providing the option to comment and give feedback to the different tasks.

The on-site teams Alpha and Beta did QA on the product delivered by the Gamma team. In order for the teams to do QA they had to follow a set of guidelines to ensure that the quality of the delivered product was of a certain level.

Open Work Area was practiced in all three teams. The on-site members of team Gamma were seated in the open work area in the close vicinity to both team Alpha and Beta. The open area enabled faster coordination across the teams because team members got hold of each other instantly.

The teams used various communication tools to instantly message each other. The tools facilitated coordination on a frequent basis and provided an important platform for communication and collaboration between the teams.

3.2 Feedback

The coordination mechanisms of type feedback is divided into *personal* and *group* mode. In the *personal mode* the individual serves as the mechanism and this mode is further divided into vertical (communication with a manager) and horizontal (communication with other team members). *Group mode* involves a group of individuals, and is divided into scheduled and unscheduled meetings. This division is mainly in order to differentiate between the more routine encounters and the informal conversations between co-workers [12].

Informal ad hoc conversations, between two or more peo-

ple, typically happened by the team members’ seats. This coordination mechanism was the one that happened most frequently and was regarded as very valuable by the team members. The mechanism also facilitated quick coordination and decision-making because the conversations took place instantly. This is illustrated by the following statement by a business developer:

“I think to make the communication more informal and have less regularly scheduled meetings is perhaps the way to go.”

A developer also commented that informal ad hoc conversations were preferred:

“I simply just walk over and talk, it’s more effective than writing a Jira issue with the overhead it entails. As a consequence I have very few e-mails, very few meetings, and that’s great.”

Every morning Gamma conducted a daily stand-up meeting with representatives from Alpha and Beta, therefore this meeting was a form of Scrum-of-Scrums. The first 15 minutes of the meeting was with the Alpha team, the subsequent 15 minutes was with the Beta team. The meeting was quite formal and rigid. They were organized so that the on-site team lead ran through each task in Jira so that the responsible developer could give status on the task. The meeting lead to little coordination, but was more of a status update, as one developer commented:

“I think there’s too little discussions regarding obstacles, it’s more like giving a status report.”

Furthermore, both Alpha and Beta had their own stand-up meetings with their respective team members, these meetings were more informal. Because Alpha was distributed with two developers off-site their stand-up meetings were conducted using video. However, the meetings in Alpha were as valuable as those in Beta even though the meetings in Alpha were distributed, where the use of video was a positive contributor. Table 3 shows an overview of what roles participated in the different daily stand-up meetings. Some roles had to attend several daily stand-up meetings every day. For example, the Tech Liaison who attended three daily stand-up meetings, as well as the whole Gamma team which was present at two daily stand-up meetings. The

Table 3: Roles participating in the different team’s daily stand-up meetings.

Roles	Teams			
	A	B	A & G	B & G
Team Alpha:				
Project Manager	X			
Team Lead	X			
Tech Liaison	X		X	X
Developers	X			
Team Beta:				
Project Manager		X		
Team Lead		X		
Test Manager		X		X
Developers		X		
Team Gamma:				
Project Manager			X	X
Team Lead			X	X
Developers			X	X

purpose of the Tech Liaison role was to create and maintain consistency across teams and across the technical platform, in which the role was considered valuable and important in a large-scale perspective.

The goal of the Retrospective was to improve the way the teams worked together. The meeting therefore included a review of what they did well, and what they could improve. All members from Gamma were present, as well as members from the relevant Norwegian team. This meeting ensured a continuous improvement of the inter-team coordination mechanisms.

At the Demo meeting, Gamma demonstrated the product they had developed with members such as the technical domain expert, test lead and business representatives, from the Norwegian team that owned that specific product. The demo facilitated coordination because it provided an arena for creating common expectations and understanding the finished product.

At the Sprint Planning meeting team Gamma selected the tasks they were going to develop in the upcoming sprint. In that way the meeting ensured the focus of the next sprint. This decision changed rarely, close to never, during the sprint. The sprint planning meeting ensured coordination between the teams because that was where the tasks were allocated amongst the members in team Gamma, based on the priorities made by the Norwegian team that was responsible for that specific area.

The purpose of the Backlog grooming was to ensure that Gamma had a thorough understanding of the new business requirements. The meeting therefore consisted of a presentation of the business requirements provided by a business representative, and a run through of technical recommendations given by the technical domain expert, both from the Norwegian team that was responsible for that specific area. The meeting ensured coordination between the teams because that was where Gamma was provided with in-depth information about the requirements.

4. DISCUSSION AND CONCLUSION

In this section we discuss our research question: What coordination mechanisms facilitate large-scale agile development?

In a study of 12 co-located development teams, Dingsøy et al. [2] reported 14 inter-team coordination mechanism. We found that 11 inter-team coordination mechanisms was present in the project. In large-scale settings, one would expect that meetings provide the best opportunity for coordination. In our study, informal ad hoc conversations were more important and also more frequently used. Our findings are consonant with those of Dingsøy et al. [2], who found that informal communication was the most important mechanism. Informal communication was supported by the open work area (ibid). Even though informal ad hoc conversations in our study facilitated coordination between the teams in an efficient way and led to quick problem-solving, a negative aspect is that it was dependent on each individual. In this program, the ad hoc conversations were necessary because the scheduled meetings were more about reporting status than coordinating work. Individuals in large-scale projects will not have necessary knowledge regarding the whole system, therefore they need to rely on their network for solving complex tasks [15]. However, some people are outgoing and know many people and will naturally contact other teams while others are more prone to working alone. For example, new team members do not necessarily know who to ask and will benefit more from scheduled meetings. Another benefit of using scheduled meetings as a coordination mechanism is that everyone is involved and receive the same information.

There was little coordination of tasks in the daily stand-up meetings. While we would expect that the daily stand-up meetings with two teams ("scrum of scrum-meetings") would have more coordination than the local stand-up meetings, the opposite was the case. However, there was little coordination in both types of meetings. Our findings support an earlier study of daily stand-up meetings which found that coordination constituted only 7% of the meeting [16]. Additionally, if they had organized the stand-up by letting people speak in turns instead of focusing on the tasks in Jira, the meeting might have led to more coordination because it would have had less focus on reporting status on each of the tasks.

The framework of Van de Ven et al. [12] was valuable as a tool to map coordination mechanisms in a large-scale program. In particular, it was valuable in terms of mapping impersonal modes of coordination such as rules, plans and communication systems (programming), because the framework makes you aware of what to look for and to understand the concept of coordination. These coordination mechanisms are usually harder to uncover because the researcher has to specifically ask for them. While meetings are visible, programming type coordination mechanisms are written rules and guidelines that are followed. It is more difficult to know that a team member works a certain way because of these guidelines, they may have turned into norms and routines several months earlier.

Norms are essential in influencing how agile team members work [17]. For example, the norm in the daily stand-up meetings was that the team lead asked for status for each of the tasks in the sprint, which resulted in little coordination in the meetings. Agile teams should incorporate norms and routines that facilitate coordination in all the different arenas. Most of the coordination in the program happened in backlog grooming and sprint planning meetings. In these meetings, tasks were assigned to each of the team members and rarely changed during the sprint. Coordination of de-

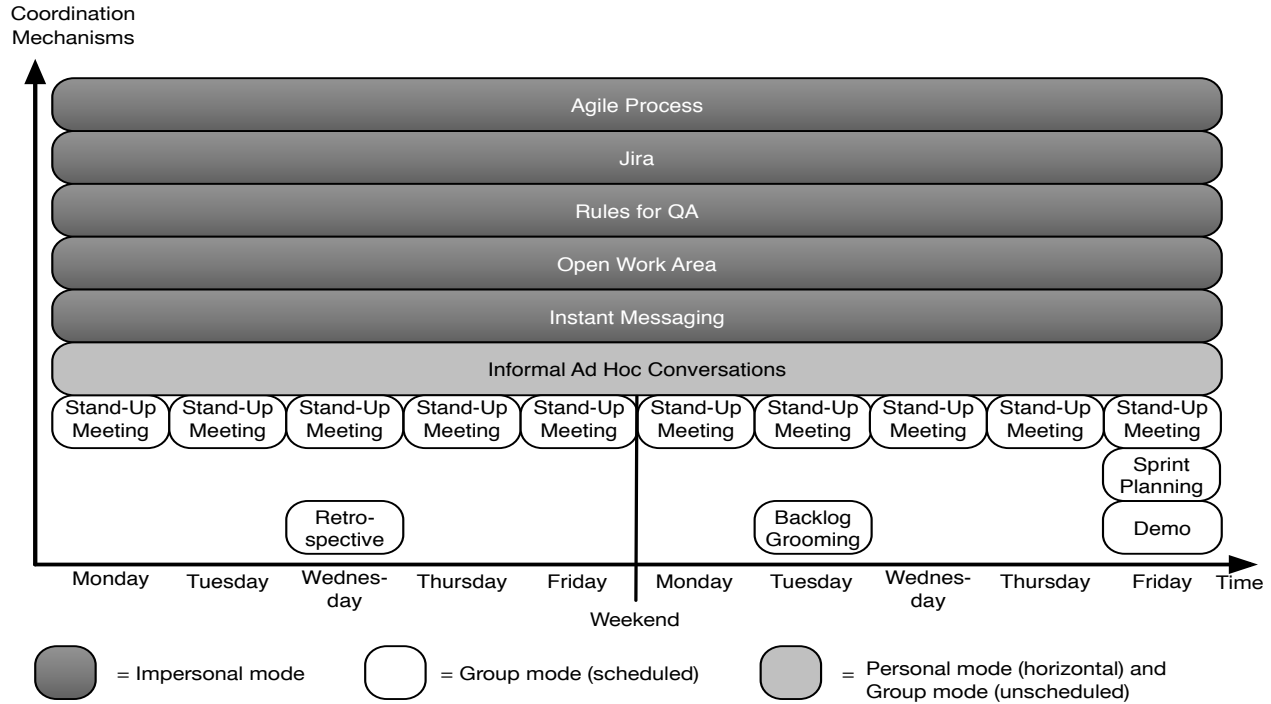


Figure 3: The coordination mechanisms when they were present during a sprint.

dependencies between tasks mostly happened through informal ad hoc conversations and by instant-messaging.

In large scale agile development, many practitioners create new roles, such as "Chief Product Owner" (CPO) in order to have someone responsible for the end product and the inter-team coordination [11]. In our study they created a role called Tech Liaison. The individual having this role possessed the technical insights into the entire product portfolio and served as a link between the different teams. The role was therefore particularly valuable for coordinating teams and facilitating the large-scale development.

In addition, many practitioners also conduct daily stand-up meetings/Scrum of scrums with participants from more than one team. However, in this case, those meetings did not lead to effective coordination. The structure of the meeting was a rigorous run-through of the current Jira tasks, which lead to a reporting of status instead of actual coordination. This style of reporting status on each Jira task gave rather the impression that the project manager on-site wanted control over the work performed off-site, rather than coordinating themselves. However, the use of video was perceived as contributing positively to the meeting.

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